

Code Quality & Security Audit Report

Comprehensive Static Code & Dependency Analysis

PROJECT	DART SDK	TARGET PLATFORMS	STATE / NETWORK
skills_check	^3.6.0	android, ios, web	setState / dio

1. Executive Summary & Category Breakdown

CATEGORY	HIGH SEVERITY	MEDIUM SEVERITY	LOW SEVERITY	TOTAL ISSUES
Security & Secrets	23	0	0	23
Performance & UI Lag	1	0	0	1
Memory & Lifecycle Leaks	40	0	0	40
Maintainability & Class Size	0	24	0	24
Total Detected Issues	64	24	0	88

2. Critical Security Findings

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
SEC-SEC-001	CRITICAL	ios/Runner/GoogleService-Info_dev.plist	6	Hardcoded API Key: Hardcoded credential or API secret pattern detected in ios/Runner/GoogleService-Info_dev.plist at line 6. <i>Risk: Storing credentials in clear text within code or resource files can result in security breaches if the code is decompiled or leaked.</i>
SEC-SEC-001	CRITICAL	lib/firebase_options.dart	41	Hardcoded API Key: Hardcoded credential or API secret pattern detected in lib/firebase_options.dart at line 41. <i>Risk: Storing credentials in clear text within code or resource files can result in security breaches if the code is decompiled or leaked.</i>
SEC-SEC-001	CRITICAL	lib/firebase_options.dart	49	Hardcoded API Key: Hardcoded credential or API secret pattern detected in lib/firebase_options.dart at line 49. <i>Risk: Storing credentials in clear text within code or resource files can result in security breaches if the code is decompiled or leaked.</i>
SEC-STR-001	HIGH	lib/utls/helpers/practice_persistence_service.dart	39	Insecure Token Storage: Detected storage write operation storing key "\${assessmentid}_\${modulekey}_\${modulekey}" in unencrypted storage (SharedPreferences, Hive, or local DB) at line 39. <i>Risk: Storing authentication tokens, credentials, or session keys in plain text allows malicious physical attackers with access to the device to compromise the user account.</i>
SEC-ATH-001	HIGH	lib/utls/common/constants.dart	25	Potential Hardcoded Password or Key: Found variable assignment indicating hardcoded password or authentication credential line 25 in lib/utls/common/constants.dart. <i>Risk: Hardcoding passwords in code makes them accessible to any attacker who extracts the application binary.</i>
SEC-ATH-001	HIGH	lib/utls/common/constants.dart	27	Potential Hardcoded Password or Key: Found variable assignment indicating hardcoded password or authentication credential line 27 in lib/utls/common/constants.dart.

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				<i>Risk: Hardcoding passwords in code makes them accessible to any attacker who executes the application binary.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	36	Potential Hardcoded Password or Key: Found variable assignment indicating hardcoded password or auth credential in line 36 in lib/utils/common/constants.dart. <i>Risk: Hardcoding passwords in code makes them accessible to any attacker who executes the application binary.</i>
SEC-ATH-001	HIGH	lib/utils/helpers/otp_persistence_helper.dart	53	Potential Hardcoded Password or Key: Found variable assignment indicating hardcoded password or auth credential in line 53 in lib/utils/helpers/otp_persistence_helper.dart. <i>Risk: Hardcoding passwords in code makes them accessible to any attacker who executes the application binary.</i>
SEC-ATH-001	HIGH	lib/utils/helpers/shared_pref_helper.dart	19	Potential Hardcoded Password or Key: Found variable assignment indicating hardcoded password or auth credential in line 19 in lib/utils/helpers/shared_pref_helper.dart. <i>Risk: Hardcoding passwords in code makes them accessible to any attacker who executes the application binary.</i>
SEC-FB-001	CRITICAL	pubspec.yaml	1	Overly Permissive Firebase Rules: Firebase rules could not be verified as secure. Rules (e.g. firestore.rules) were found locally. <i>Risk: Without locally declared security rules, the application cannot verify if the remote database is protected against unauthorized read/write requests.</i>
SEC-WV-001	HIGH	lib/screens/student_module/course/course_overview/course_overview.dart	661	Unsafe WebView Configuration: Detected a WebView instantiated with unrestricted JavaScript execution enabled at line 661. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-WV-001	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	2275	Unsafe WebView Configuration: Detected a WebView instantiated with unrestricted JavaScript execution enabled at line 2275. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-WV-002	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	2452	Unsafe WebView Configuration: A WebView was instantiated without a navigationDelegate to filter loaded content at line 2452. <i>Risk: Without URL checking, users or robots can navigate the WebView to arbitrary external domains, which might steal sensitive information or impersonate interfaces.</i>
SEC-WV-001	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_format_page.dart	237	Unsafe WebView Configuration: Detected a WebView instantiated with unrestricted JavaScript execution enabled at line 237. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
SEC-WV-001	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_format_page.dart	314	Unsafe WebView Configuration: A WebView was instantiated with unrestricted JavaScript execution enabled at line 314. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-WV-002	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_format_page.dart	495	Unsafe WebView Configuration: A WebView was instantiated without a navigationDelegate to filter loaded content at line 495. <i>Risk: Without URL checking, users or remote links can navigate the WebView to arbitrary external domains, which might steal to impersonate interfaces.</i>
SEC-WV-002	HIGH	lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_format_page.dart	519	Unsafe WebView Configuration: A WebView was instantiated without a navigationDelegate to filter loaded content at line 519. <i>Risk: Without URL checking, users or remote links can navigate the WebView to arbitrary external domains, which might steal to impersonate interfaces.</i>
SEC-WV-001	HIGH	lib/screens/student_module/course/course_contents/office_file_viewer.dart	32	Unsafe WebView Configuration: A WebView was instantiated with unrestricted JavaScript execution enabled at line 32. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-WV-001	HIGH	lib/screens/student_module/course/course_contents/scrom_screen.dart	41	Unsafe WebView Configuration: A WebView was instantiated with unrestricted JavaScript execution enabled at line 41. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-WV-001	HIGH	lib/screens/student_module/my_content/viewers/doc_content_viewer.dart	37	Unsafe WebView Configuration: A WebView was instantiated with unrestricted JavaScript execution enabled at line 37. <i>Risk: Enabling JavaScript in WebViews loaded pages to execute code. If untrusted remote content is displayed, it could lead to Cross-Site Scripting (XSS) attacks compromising local app data or bridge.</i>
SEC-AND-002	HIGH	android/app/src/main/AndroidManifest.xml	13	Android Insecure Configuration: The Android Manifest configures the app to allow cleartext HTTP traffic at line 13. <i>Risk: Cleartext traffic is vulnerable to interception and modification via Man-in-the-Middle attacks.</i>
SEC-AND-003	HIGH	android/app/src/main/AndroidManifest.xml	17	Android Insecure Configuration: An Android component is marked exported="true" at line 17 in android/app/src/main/AndroidManifest.xml. <i>Risk: Exported activities can be started by other application on the device. If they receive arguments or intents without validation, they could lead to intent injection or unauthorized access.</i>

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
SEC-IOS-001	HIGH	ios/Runner/Info.plist	29	iOS Insecure Configuration: The Info.plist contains NSAllowsArbitraryLoads at line 29. <i>Risk: Enabling arbitrary loads disables App Transport Security (ATS) checks for the app, permitting unencrypted connections (HTTP) to any server.</i>

- Security Remediation Action Items
- Remove secret and rotate credential
 - Use platform secure storage
 - Retrieve passwords dynamically from user input or encrypted vaults.
 - Apply least-privilege authenticated access
 - Restrict navigation, sources and JS capabilities
 - Harden release configuration
 - Minimize exceptions and harden release settings

3. High-Priority Performance Flaws: Controller in build()

Instantiating controllers inside build() forces memory re-allocations on every frame render, causing UI stutter, dropped text inputs, and lost scroll states.

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
Feedback Screen	lib/screens/student_module/feedback/feedback_screen.dart	132	Controller Instantiation Inside Widget build() Method

Standard Fix Architecture:

```
class _SafeInputWidgetState extends State<SafeInputWidget> {
  late final TextEditingController _controller;

  @override
  void initState() {
    super.initState();
    _controller = TextEditingController(); // Allocated once
  }

  @override
  void dispose() {
    _controller.dispose(); // Safely freed from memory
    super.dispose();
  }

  @override
  Widget build(BuildContext context) => TextField(controller: _controller);
}
```

4. Memory & Resource Leaks

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-001 (HIGH)	lib/utils/helpers/notification_service.dart	105	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/utils/helpers/notification_service.dart	112	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/utils/helpers/connectivity.dart	28	Unclosed StreamSubscription
MEM-002 (HIGH)	lib/reusable_component/question_widgets/assessment_audio_answer_field.dart	186	Unclosed Timer
MEM-002 (HIGH)	lib/reusable_component/circular_timer/circular_countdown_timer.dart	91	Unclosed Timer
MEM-001 (HIGH)	lib/screens/student_module/student_login/login_with_auth_code.dart	25	Unclosed StreamSubscription

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-001 (HIGH)	lib/screens/student_module/student_login/login.dart	34	Unclosed StreamSubscription
MEM-002 (HIGH)	lib/screens/student_module/student_login/otp_Screen.dart	57	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/student_login/otp_Screen.dart	62	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/student_login/otp_Screen.dart	70	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/student_login/otp_Screen.dart	106	Unclosed Timer
MEM-003 (HIGH)	lib/screens/student_module/student_login/verify_otp_bloc.dart	28	AnimationController Not Disposed
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	96	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	99	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	102	Unclosed StreamSubscription
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/youtube_player_screen.dart	102	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/youtube_player_screen.dart	425	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/images_content.dart	72	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/images_content.dart	412	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/video_player_screen.dart	121	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/video_player_screen.dart	799	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/video_player_screen.dart	818	Unclosed Timer
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	125	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	130	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	135	Unclosed StreamSubscription
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	98	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	468	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/audio_player_screen.dart	486	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/text_content.dart	249	Unclosed Timer
MEM-002 (HIGH)	lib/screens/student_module/course/course_contents/text_content.dart	268	Unclosed Timer
MEM-001 (HIGH)	lib/screens/student_module/assessment/assessments_test/answer_sheet/assessment_answer_sheet.dart	1017	Unclosed StreamSubscription
MEM-002 (HIGH)	lib/screens/student_module/assessment/assessments_test/widgets/assessment_audio_answer_field.dart	140	Unclosed Timer
MEM-001 (HIGH)	lib/screens/student_module/feedback/feedback_screen.dart	40	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/my_space/my_space_screen.dart	35	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/student_module/dashboard/widgets/home_screen.dart	46	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/support/support_screen.dart	65	Unclosed StreamSubscription

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-001 (HIGH)	lib/screens/parent_module/parent_profile/parent_personal_info_screen.dart	53	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/parent_module/parent_profile/parent_personal_info_screen.dart	61	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart	37	Unclosed StreamSubscription
MEM-001 (HIGH)	lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart	50	Unclosed StreamSubscription

5. Class Bloat & Technical Debt (Files > 500 LOC)

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
lib/module/get_assessment_details_by_id_model.dart	816	General	Split large files into smaller focused units
lib/module/assessment_progress_model.dart	579	General	Split large files into smaller focused units
lib/module/course_details_model.dart	1007	General	Split large files into smaller focused units
lib/utils/theme/style.dart	573	Utility Helpers	Split large files into smaller focused units
lib/reusable_component/question_widgets/mcq_option.dart	529	General	Split large files into smaller focused units
lib/screens/student_module/course/course_navigation_router.dart	694	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_overview/course_overview.dart	707	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/epub_screen.dart	646	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart	3370	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_format_page.dart	4013	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/document_content/documents_content.dart	939	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/document_content/document_content_bloc.dart	659	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/video_player_screen.dart	839	UI / View	Split large files into smaller focused units
lib/screens/student_module/course/course_contents/audio_player_screen.dart	513	UI / View	Split large files into smaller focused units
lib/screens/student_module/assessment/assessments_test/assessments_test_question_bloc.dart	1443	UI / View	Split large files into smaller focused units
lib/screens/student_module/assessment/assessments_test/answer_sheet/assessment_answer_sheet.dart	1137	UI / View	Split large files into smaller focused units
lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart	2099	UI / View	Split large files into smaller focused units
lib/screens/student_module/assessment/assessment_module/module_screen.dart	601	UI / View	Split large files into smaller focused units
lib/screens/student_module/my_space/myspace_modules.dart	961	UI / View	Split large files into smaller focused units

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart	517	UI / View	Split large files into smaller focused units

6. Unresolved Production TODOs

FILE PATH	LINE	MISSING FEATURE / INCOMPLETE CODE
lib/reusable_component/add_child_bottom_sheet.dart	292	Found unresolved developer note in lib/reusable_component/add_child_bottom_sheet.dart at line 292: Navigate to Consent Policy
lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart	1060	Found unresolved developer note in lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart at line 1060: Handle this case.
lib/screens/parent_module/parent_login/link_child_screen.dart	348	Found unresolved developer note in lib/screens/parent_module/parent_login/link_child_screen.dart at line 348: Navigate to Consent Policy
lib/screens/parent_module/parent_login/parent_login_screen.dart	327	Found unresolved developer note in lib/screens/parent_module/parent_login/parent_login_screen.dart at line 327: Navigate to Terms & Conditions

7. Recommended Action Plan

- Phase 1 — Hotfix (Days 1–2):
- Remove secret and rotate credential
 - Remove secret and rotate credential
 - Remove secret and rotate credential
- Phase 2 — Performance & Memory (Days 3–5):
- Convert the widget to a StatefulWidget, instantiate TabController in initState(), and dispose it in dispose().
 - Cancel subscriptions in dispose/close
 - Cancel subscriptions in dispose/close
- Phase 3 — Refactoring & Debt (Week 2):
- Split large files into smaller focused units
 - Split large files into smaller focused units
 - Address pending TODO comments in lib/reusable_component/add_child_bottom_sheet.dart at line 292.

8. Master Claude AI Fix Command (Automated Batch Fix)

Pass the prompt below to Claude to automatically apply non-breaking fixes for all identified findings:

🛠 Master Fix Command for Claude

Please review and resolve the following 88 audit finding(s) in the project.

CRITICAL: Preserve all existing functionality, imports, styles, and public APIs while applying these fixes.

📁 File: `ios/Runner/GoogleService-Info\_dev.plist` (1 issue)

1. [SECURITY - CRITICAL] Hardcoded API Key (Line 6)

- **Evidence:** `<string>AIza...[MASKED]...nSD0</string>`

- **Fix Instruction:**

...
// In ios/Runner/GoogleService-Info_dev.plist (Line 6):
// Move hardcoded credential into environment variables or secret store.

📁 File: `lib/firebase\_options.dart` (2 issues)

2. [SECURITY - CRITICAL] Hardcoded API Key (Line 41)

- **Evidence:** `apiK...[MASKED]...3kg`,`

- **Fix Instruction:**

...
// In lib/firebase_options.dart (Line 41):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

...
// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

...
// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');

3. [SECURITY - CRITICAL] Hardcoded API Key (Line 49)

- **Evidence:** `apiK...[MASKED]...SD0`,`

- **Fix Instruction:**

...
// In lib/firebase_options.dart (Line 49):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

...
// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

...
// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');

📁 File: `lib/utils/helpers/practice\_persistence\_service.dart` (1 issue)

4. [SECURITY - HIGH] Insecure Token Storage (Line 39)

- **Evidence:** `await \_completedBox.put('\${assessmentId}\_\${moduleKey}\_\${partId}', true);`

- **Fix Instruction:**

...
// In lib/utils/helpers/practice_persistence_service.dart (Line 39):
// Replace plaintext SharedPreferences / unencrypted storage with FlutterSecureStorage:
import 'package:flutter_secure_storage/flutter_secure_storage.dart';

...
final secureStorage = const FlutterSecureStorage();
// Write: await secureStorage.write(key: 'auth_token', value: token);
// Read: final token = await secureStorage.read(key: 'auth_token');

📁 File: `lib/utils/common/constants.dart` (3 issues)

5. [SECURITY - HIGH] Potential Hardcoded Password or Auth Key (Line 25)

- **Evidence:** `static const String resetPassword = '/sc/v1/student/reset-password';`

- **Fix Instruction:**

...
// In lib/utils/common/constants.dart (Line 25):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

...
// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

...
// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();

```

final apiKey = await storage.read(key: 'api_secret_key');
'''

### 6. [SECURITY - HIGH] Potential Hardcoded Password or Auth Key (Line 27)
- **Evidence:** `static const String requestPasswordReset = '/sc/v1/student/request-password-reset';`
- **Fix Instruction:**
'''
// In lib/utils/common/constants.dart (Line 27):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');
'''

### 7. [SECURITY - HIGH] Potential Hardcoded Password or Auth Key (Line 36)
- **Evidence:** `static const String refreshAccessToken = '/sc/v1/student/refresh-access-token';`
- **Fix Instruction:**
'''
// In lib/utils/common/constants.dart (Line 36):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');
'''

## 📁 File: `lib/utils/helpers/otp_persistence_helper.dart` (1 issue)

### 8. [SECURITY - HIGH] Potential Hardcoded Password or Auth Key (Line 53)
- **Evidence:** `static const String _key = 'pending_otp_data';`
- **Fix Instruction:**
'''
// In lib/utils/helpers/otp_persistence_helper.dart (Line 53):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');
'''

## 📁 File: `lib/utils/helpers/shared_pref_helper.dart` (1 issue)

### 9. [SECURITY - HIGH] Potential Hardcoded Password or Auth Key (Line 19)
- **Evidence:** `final String refreshToken = 'refreshToken';`
- **Fix Instruction:**
'''
// In lib/utils/helpers/shared_pref_helper.dart (Line 19):
// Step 1: Remove hardcoded credential and store it in environment config or secure storage.
// Step 2: Use flutter_dotenv or flutter_secure_storage:

// Option A: Using flutter_dotenv (.env file):
// Add key to .env: API_SECRET_KEY=your_value_here
import 'package:flutter_dotenv/flutter_dotenv.dart';
final apiKey = dotenv.env['API_SECRET_KEY'] ?? '';

// Option B: Using flutter_secure_storage (for runtime sensitive tokens):
import 'package:flutter_secure_storage/flutter_secure_storage.dart';
final storage = const FlutterSecureStorage();
final apiKey = await storage.read(key: 'api_secret_key');
'''

## 📁 File: `pubspec.yaml` (1 issue)

### 10. [SECURITY - CRITICAL] Overly Permissive Firebase Rules (Line 1)
- **Evidence:** `Firebase dependencies declared without local security rules`
- **Fix Instruction:**
'''

```

```
// In firestore.rules:
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /{document=**} {
      allow read, write: if request.auth != null;
    }
  }
}
...
## File: `lib/screens/student_module/course/course_overview/course_overview.dart` (2 issues)

### 11. [SECURITY - HIGH] Unsafe WebView Configuration (Line 661)
- **Evidence:** `..setJavaScriptMode(JavaScriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_overview/course_overview.dart (Line 661):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
),
)
...

### 12. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 707 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_overview/course_overview.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart` (6 issues)

### 13. [SECURITY - HIGH] Unsafe WebView Configuration (Line 2275)
- **Evidence:** `..setJavaScriptMode(JavaScriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart (Line 2275):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
),
)
...

### 14. [SECURITY - HIGH] Unsafe WebView Configuration (Line 2452)
- **Evidence:** `print('Loaded page $index into WebView (${finalHtml.length} chars)');`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart (Line 2452):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
),
)
...

### 15. [MEMORY - HIGH] Unclosed StreamSubscription (Line 96)
- **Evidence:** `_audioPlayer!.onPlayerStateChanged.listen((state) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart (Line 96):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer!.onPlayerStateChanged.listen((state) {;

// 3. Clean up in dispose() or onClose():
@override
```

```

void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 16. [MEMORY - HIGH] Unclosed StreamSubscription (Line 99)
- **Evidence:** ` _audioPlayer!.onPositionChanged.listen((pos) {}`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart (Line 99):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer!.onPositionChanged.listen((pos) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 17. [MEMORY - HIGH] Unclosed StreamSubscription (Line 102)
- **Evidence:** ` _audioPlayer!.onDurationChanged.listen((dur) {}`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart (Line 102):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer!.onDurationChanged.listen((dur) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 18. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 3370 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_page.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## File: `lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart` (5 issues)

### 19. [SECURITY - HIGH] Unsafe WebView Configuration (Line 237)
- **Evidence:** `..setJavaScriptMode(JavascriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart (Line 237):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate({
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
})
...

### 20. [SECURITY - HIGH] Unsafe WebView Configuration (Line 314)
- **Evidence:** `..setJavaScriptMode(JavascriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart (Line 314):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate({
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
})
...

```

```
### 21. [SECURITY - HIGH] Unsafe WebView Configuration (Line 495)
- **Evidence:** `print('Loaded page $index into WebView (${finalHtml.length} chars)');`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart (Line 495):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
)
...

### 22. [SECURITY - HIGH] Unsafe WebView Configuration (Line 519)
- **Evidence:** `Unloaded page $i from WebView (distance=$distance from $centerPage);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart (Line 519):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
)
...

### 23. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 4013 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/epubx_reader_new_formate_page.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/student_module/course/course_contents/office_file_viewer.dart` (1 issue)

### 24. [SECURITY - HIGH] Unsafe WebView Configuration (Line 32)
- **Evidence:** `..setJavaScriptMode(JavascriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/office_file_viewer.dart (Line 32):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
)
...
## File: `lib/screens/student_module/course/course_contents/scrom_screen.dart` (1 issue)

### 25. [SECURITY - HIGH] Unsafe WebView Configuration (Line 41)
- **Evidence:** `..setJavaScriptMode(JavascriptMode.unrestricted)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/scrom_screen.dart (Line 41):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate(
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
)
...
## File: `lib/screens/student_module/my_content/viewers/doc_content_viewer.dart` (1 issue)

### 26. [SECURITY - HIGH] Unsafe WebView Configuration (Line 37)
- **Evidence:** `..setJavaScriptMode(JavascriptMode.unrestricted)`
- **Fix Instruction:
```

```
// In lib/screens/student_module/my_content/viewers/doc_content_viewer.dart (Line 37):
// Restrict JavaScript and validate loaded URLs with navigationDelegate:
NavigationDelegate({
  onNavigationRequest: (NavigationRequest request) {
    final uri = Uri.parse(request.url);
    if (uri.scheme == 'https' && uri.host.endsWith('yourdomain.com')) {
      return NavigationDecision.navigate;
    }
    return NavigationDecision.prevent;
  },
},
);
```

File: `android/app/src/main/AndroidManifest.xml` (2 issues)

27. [SECURITY - HIGH] Android Insecure Configuration (Line 13)

```
- **Evidence:** `android:usesCleartextTraffic="true">`
- **Fix Instruction:**
...
<!-- In android/app/src/main/AndroidManifest.xml (Line 13) -->
<!-- Disable cleartext HTTP traffic or use network_security_config -->
<application
  android:usesCleartextTraffic="false"
  ... >
```

28. [SECURITY - HIGH] Android Insecure Configuration (Line 17)

```
- **Evidence:** `android:exported="true"`
- **Fix Instruction:**
...
<!-- In android/app/src/main/AndroidManifest.xml (Line 17) -->
<!-- Set exported="false" for internal activities or protect with permissions -->
<activity
  android:name="..."
  android:exported="false">
</activity>
```

File: `ios/Runner/Info.plist` (1 issue)

29. [SECURITY - HIGH] iOS Insecure Configuration (Line 29)

```
- **Evidence:** `<key>NSAllowsArbitraryLoads</key>`
- **Fix Instruction:**
...
<!-- In ios/Runner/Info.plist (Line 29) -->
<!-- Disable global ATS bypass and configure domain-specific exceptions if needed -->
<key>NSAppTransportSecurity</key>
<dict>
  <key>NSAllowsArbitraryLoads</key>
  <false/>
</dict>
```

File: `lib/utils/helpers/notification\_service.dart` (2 issues)

30. [MEMORY - HIGH] Unclosed StreamSubscription (Line 105)

```
- **Evidence:** `FirebaseMessaging.onMessage.listen((RemoteMessage message) {}`
- **Fix Instruction:**
...
```

// In lib/utils/helpers/notification_service.dart (Line 105):

// 1. Declare subscription variable in the class/State:

```
StreamSubscription? _subscription;
```

// 2. Assign when listening (e.g. in initState):

```
_subscription = FirebaseMessaging.onMessage.listen((RemoteMessage message) {};
```

// 3. Clean up in dispose() or onClose():

```
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
```

31. [MEMORY - HIGH] Unclosed StreamSubscription (Line 112)

```
- **Evidence:** `FirebaseMessaging.onMessageOpenedApp.listen((RemoteMessage message) {}`
- **Fix Instruction:**
...
```

// In lib/utils/helpers/notification_service.dart (Line 112):

// 1. Declare subscription variable in the class/State:

```
StreamSubscription? _subscription;
```

// 2. Assign when listening (e.g. in initState):

```
_subscription = FirebaseMessaging.onMessageOpenedApp.listen((RemoteMessage message) {};
```

// 3. Clean up in dispose() or onClose():

```
@override
void dispose() {
  _subscription?.cancel();
}
```

```

super.dispose();
}
...
## 📄 File: `lib/Utils/helpers/connectivity.dart` (1 issue)

### 32. [MEMORY - HIGH] Unclosed StreamSubscription (Line 28)
- **Evidence:** `.listen((List<ConnectivityResult> connectivityResult) {`
- **Fix Instruction:**
...
// In lib/Utils/helpers/connectivity.dart (Line 28):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = .listen((List<ConnectivityResult> connectivityResult) {;

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## 📄 File: `lib/reusable_component/question_widgets/assessment_audio_answer_field.dart` (1 issue)

### 33. [MEMORY - HIGH] Unclosed Timer (Line 186)
- **Evidence:** `CircularCountDownTimer(`
- **Fix Instruction:**
...
// In lib/reusable_component/question_widgets/assessment_audio_answer_field.dart (Line 186):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = CircularCountDownTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...
## 📄 File: `lib/reusable_component/circular_timer/circular_countdown_timer.dart` (1 issue)

### 34. [MEMORY - HIGH] Unclosed Timer (Line 91)
- **Evidence:** `const CircularCountDownTimer({`
- **Fix Instruction:**
...
// In lib/reusable_component/circular_timer/circular_countdown_timer.dart (Line 91):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = const CircularCountDownTimer({;

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...
## 📄 File: `lib/screens/student_module/student_login/login_with_auth_code.dart` (1 issue)

### 35. [MEMORY - HIGH] Unclosed StreamSubscription (Line 25)
- **Evidence:** `_loginBloc.loginState.listen((state) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/login_with_auth_code.dart (Line 25):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _loginBloc.loginState.listen((state) {;

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## 📄 File: `lib/screens/student_module/student_login/login.dart` (1 issue)

```

```

### 36. [MEMORY - HIGH] Unclosed StreamSubscription (Line 34)
- **Evidence:** `_loginBloc.loginState.listen((state) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/login.dart (Line 34):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _loginBloc.loginState.listen((state) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## File: `lib/screens/student_module/student_login/otp_Screen.dart` (4 issues)

### 37. [MEMORY - HIGH] Unclosed Timer (Line 57)
- **Evidence:** `_initializeTimer();`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/otp_Screen.dart (Line 57):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _initializeTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 38. [MEMORY - HIGH] Unclosed Timer (Line 62)
- **Evidence:** `Future<void> _initializeTimer() async {`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/otp_Screen.dart (Line 62):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = Future<void> _initializeTimer() async {;

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 39. [MEMORY - HIGH] Unclosed Timer (Line 70)
- **Evidence:** `_bloc.startTimer(initialStart: initialStart);`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/otp_Screen.dart (Line 70):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(initialStart: initialStart);

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 40. [MEMORY - HIGH] Unclosed Timer (Line 106)
- **Evidence:** `startTimer();`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/otp_Screen.dart (Line 106):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = startTimer();

```



```
// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

## 📄 File: `lib/screens/student_module/student_login/verify_otp_bloc.dart` (1 issue)

### 41. [MEMORY - HIGH] AnimationController Not Disposed (Line 28)
- **Evidence:** `final TextEditingController otpCtr = TextEditingController();`
- **Fix Instruction:**
...
// In lib/screens/student_module/student_login/verify_otp_bloc.dart (Line 28):
// 1. Declare as a state field:
late final TextEditingController _textController;

// 2. Initialize in initState():
@override
void initState() {
  super.initState();
  _textController = TextEditingController();
}

// 3. Always dispose in dispose():
@override
void dispose() {
  _textController.dispose();
  super.dispose();
}
...

## 📄 File: `lib/screens/student_module/course/course_contents/youtube_player_screen.dart` (2 issues)

### 42. [MEMORY - HIGH] Unclosed Timer (Line 102)
- **Evidence:** `_bloc.setAndStartTimer(`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/youtube_player_screen.dart (Line 102):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.setAndStartTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 43. [MEMORY - HIGH] Unclosed Timer (Line 425)
- **Evidence:** `_bloc.startTimer(widget.part, context);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/youtube_player_screen.dart (Line 425):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(widget.part, context);

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

## 📄 File: `lib/screens/student_module/course/course_contents/images_content.dart` (2 issues)

### 44. [MEMORY - HIGH] Unclosed Timer (Line 72)
- **Evidence:** `_bloc.setAndStartTimer(`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/images_content.dart (Line 72):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.setAndStartTimer();

// 3. Cancel inside dispose() or onClose():
@override
```

```

void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 45. [MEMORY - HIGH] Unclosed Timer (Line 412)
- **Evidence:** `_bloc.startTimer(widget.part, context);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/images_content.dart (Line 412):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(widget.part, context);;

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

## File: `lib/screens/student_module/course/course_contents/video_player_screen.dart` (4 issues)

### 46. [MEMORY - HIGH] Unclosed Timer (Line 121)
- **Evidence:** `_bloc.setAndStartTimer(`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/video_player_screen.dart (Line 121):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.setAndStartTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 47. [MEMORY - HIGH] Unclosed Timer (Line 799)
- **Evidence:** `_bloc.pauseTimer();`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/video_player_screen.dart (Line 799):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.pauseTimer();;

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 48. [MEMORY - HIGH] Unclosed Timer (Line 818)
- **Evidence:** `_bloc.startTimer(widget.part, context);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/video_player_screen.dart (Line 818):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(widget.part, context);;

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 49. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 839 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/video_player_screen.dart:

```

```
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## File: `lib/screens/student_module/course/course_contents/audio_player_screen.dart` (7 issues)

### 50. [MEMORY - HIGH] Unclosed StreamSubscription (Line 125)
- **Evidence:** `_audioPlayer.durationStream.listen((d) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 125):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer.durationStream.listen((d) {;

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 51. [MEMORY - HIGH] Unclosed StreamSubscription (Line 130)
- **Evidence:** `_audioPlayer.positionStream.listen((p) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 130):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer.positionStream.listen((p) {;

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 52. [MEMORY - HIGH] Unclosed StreamSubscription (Line 135)
- **Evidence:** `_audioPlayer.playerStateStream.listen((state) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 135):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _audioPlayer.playerStateStream.listen((state) {;

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 53. [MEMORY - HIGH] Unclosed Timer (Line 98)
- **Evidence:** `.setAndStartTimer(`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 98):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = .setAndStartTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 54. [MEMORY - HIGH] Unclosed Timer (Line 468)
- **Evidence:** `_bloc.pauseTimer();`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 468):
// 1. Declare Timer variable in the class/State:
```

```

Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.pauseTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 55. [MEMORY - HIGH] Unclosed Timer (Line 486)
- **Evidence:** `_bloc.startTimer(widget.part, context);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart (Line 486):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(widget.part, context);

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 56. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 513 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/audio_player_screen.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## File: `lib/screens/student_module/course/course_contents/text_content.dart` (2 issues)

### 57. [MEMORY - HIGH] Unclosed Timer (Line 249)
- **Evidence:** `_bloc.pauseTimer();`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/text_content.dart (Line 249):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.pauseTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

### 58. [MEMORY - HIGH] Unclosed Timer (Line 268)
- **Evidence:** `_bloc.startTimer(widget.part, context);`
- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/text_content.dart (Line 268):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = _bloc.startTimer(widget.part, context);

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

## File: `lib/screens/student_module/assessment/assessments_test/answer_sheet/assessment_answer_sheet.dart` (2 issues)

### 59. [MEMORY - HIGH] Unclosed StreamSubscription (Line 1017)
- **Evidence:** `_player.durationStream.listen((d) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/answer_sheet/assessment_answer_sheet.dart (Line 1017):

```

```
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _player.durationStream.listen((d) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 60. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 1137 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/answer_sheet/assessment_answer_sheet.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## File: `lib/screens/student_module/assessment/assessments_test/widgets/assessment_audio_answer_field.dart` (1 issue)

### 61. [MEMORY - HIGH] Unclosed Timer (Line 140)
- **Evidence:** `CircularCountDownTimer(`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/widgets/assessment_audio_answer_field.dart (Line 140):
// 1. Declare Timer variable in the class/State:
Timer? _timer;

// 2. Assign when creating:
_timer = CircularCountDownTimer();

// 3. Cancel inside dispose() or onClose():
@override
void dispose() {
  _timer?.cancel();
  super.dispose();
}
...

## File: `lib/screens/student_module/feedback/feedback_screen.dart` (2 issues)

### 62. [MEMORY - HIGH] Unclosed StreamSubscription (Line 40)
- **Evidence:** `aGeneralBloc.studentProfileStream.listen((profile) {`
- **Fix Instruction:**
...
// In lib/screens/student_module/feedback/feedback_screen.dart (Line 40):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = aGeneralBloc.studentProfileStream.listen((profile) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 63. [PERFORMANCE - HIGH] Controller Instantiation Inside Widget build() Method (Line 132)
- **Evidence:** `_tabController = TabController(length: availability.length, vsync: this);`
- **Fix Instruction:**
...
// In lib/screens/student_module/feedback/feedback_screen.dart (Line 132):
// Convert widget to a StatefulWidget State<YourWidget> and move controller creation out of build():
// 1. Initialize inside initState():
@override
void initState() {
  super.initState();
  // Initialize controller here once
}

// 2. Dispose in dispose():
@override
void dispose() {
  // Dispose controller here
  super.dispose();
}
...

## File: `lib/screens/student_module/my_space/my_space_screen.dart` (1 issue)
```

```
### 64. [MEMORY - HIGH] Unclosed StreamSubscription (Line 35)
- **Evidence:** `aGeneralBloc.studentProfileStream.listen((profile) {}`
- **Fix Instruction:**
...
// In lib/screens/student_module/my_space/my_space_screen.dart (Line 35):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = aGeneralBloc.studentProfileStream.listen((profile) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## File: `lib/screens/student_module/dashboard/widgets/home_screen.dart` (1 issue)

### 65. [MEMORY - HIGH] Unclosed StreamSubscription (Line 46)
- **Evidence:** `aGeneralBloc.studentProfileStream.listen((profile) {}`
- **Fix Instruction:**
...
// In lib/screens/student_module/dashboard/widgets/home_screen.dart (Line 46):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = aGeneralBloc.studentProfileStream.listen((profile) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## File: `lib/screens/support/support_screen.dart` (1 issue)

### 66. [MEMORY - HIGH] Unclosed StreamSubscription (Line 65)
- **Evidence:** `_bloc.message.listen((msg) {}`
- **Fix Instruction:**
...
// In lib/screens/support/support_screen.dart (Line 65):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _bloc.message.listen((msg) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...
## File: `lib/screens/parent_module/parent_profile/parent_personal_info_screen.dart` (2 issues)

### 67. [MEMORY - HIGH] Unclosed StreamSubscription (Line 53)
- **Evidence:** `_bloc.navigationStream.listen((shouldPop) {}`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parent_profile/parent_personal_info_screen.dart (Line 53):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _bloc.navigationStream.listen((shouldPop) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 68. [MEMORY - HIGH] Unclosed StreamSubscription (Line 61)
- **Evidence:** `_bloc.isLoading.listen((loading) {}`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parent_profile/parent_personal_info_screen.dart (Line 61):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;
```

```
// 2. Assign when listening (e.g. in initState):
_subscription = _bloc.isLoading.listen((loading) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

## 📁 File: `lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart` (3 issues)

### 69. [MEMORY - HIGH] Unclosed StreamSubscription (Line 37)
- **Evidence:** `'_parentHomeBloc.childrenDetails.listen((details) {}`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart (Line 37):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _parentHomeBloc.childrenDetails.listen((details) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 70. [MEMORY - HIGH] Unclosed StreamSubscription (Line 50)
- **Evidence:** `'_feedbackBloc.feedbackAvailability.listen((availability) {}`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart (Line 50):
// 1. Declare subscription variable in the class/State:
StreamSubscription? _subscription;

// 2. Assign when listening (e.g. in initState):
_subscription = _feedbackBloc.feedbackAvailability.listen((availability) {});

// 3. Clean up in dispose() or onClose():
@override
void dispose() {
  _subscription?.cancel();
  super.dispose();
}
...

### 71. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 517 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parents_feedback/parent_feedback_list_screen.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## 📁 File: `lib/module/get_assessment_details_by_id_model.dart` (1 issue)

### 72. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 816 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/module/get_assessment_details_by_id_model.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## 📁 File: `lib/module/assessment_progress_model.dart` (1 issue)

### 73. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 579 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/module/assessment_progress_model.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...

## 📁 File: `lib/module/course_details_model.dart` (1 issue)

### 74. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
```

```

-- **Evidence:** `Lines count: 1007 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/module/course_details_model.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/Utils/theme/style.dart` (1 issue)

### 75. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 573 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/Utils/theme/style.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/reusable_component/question_widgets/mcq_option.dart` (1 issue)

### 76. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 529 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/reusable_component/question_widgets/mcq_option.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/reusable_component/add_child_bottom_sheet.dart` (1 issue)

### 77. [CODE_QUALITY - MEDIUM] Unused Widgets/Classes/Functions (Line 292)
-- **Evidence:** `// TODO: Navigate to Consent Policy`
-- **Fix Instruction:**
...
// In lib/reusable_component/add_child_bottom_sheet.dart (Line 292):
// Complete or track pending note: "/// TODO: Navigate to Consent Policy"
// Either implement the required logic or convert it to an issue tracker ticket.
...
## 📁 File: `lib/screens/student_module/course/course_navigation_router.dart` (1 issue)

### 78. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 694 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_navigation_router.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/screens/student_module/course/course_contents/epub_screen.dart` (1 issue)

### 79. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 646 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/epub_screen.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/screens/student_module/course/course_contents/document_content/documents_content.dart` (1 issue)

### 80. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 939 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/documents_content.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## 📁 File: `lib/screens/student_module/course/course_contents/document_content/document_content_bloc.dart` (1 issue)

### 81. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
-- **Evidence:** `Lines count: 659 (Max threshold: 500)`
-- **Fix Instruction:**
...
// In lib/screens/student_module/course/course_contents/document_content/document_content_bloc.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.

```



```
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/student_module/assessment/assessments_test/assessments_test_question_bloc.dart` (1 issue)

### 82. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 1443 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/assessments_test_question_bloc.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart` (2 issues)

### 83. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 2099 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
### 84. [CODE_QUALITY - MEDIUM] Unused Widgets/Classes/Functions (Line 1060)
- **Evidence:** `// TODO: Handle this case.`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessments_test/assessments_test_question.dart (Line 1060):
// Complete or track pending note: "// TODO: Handle this case."
// Either implement the required logic or convert it to an issue tracker ticket.
...
## File: `lib/screens/student_module/assessment/assessment_module/module_screen.dart` (1 issue)

### 85. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 601 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/assessment/assessment_module/module_screen.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/student_module/my_space/myspace_modules.dart` (1 issue)

### 86. [CODE_QUALITY - MEDIUM] Large File LOC (Line 1)
- **Evidence:** `Lines count: 961 (Max threshold: 500)`
- **Fix Instruction:**
...
// In lib/screens/student_module/my_space/myspace_modules.dart:
// Break down this large file into modular components:
// 1. Extract reusable sub-widgets into separate widget files.
// 2. Move business logic, state handlers, or helpers into dedicated service or controller classes.
...
## File: `lib/screens/parent_module/parent_login/link_child_screen.dart` (1 issue)

### 87. [CODE_QUALITY - MEDIUM] Unused Widgets/Classes/Functions (Line 348)
- **Evidence:** `// TODO: Navigate to Consent Policy`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parent_login/link_child_screen.dart (Line 348):
// Complete or track pending note: "// TODO: Navigate to Consent Policy"
// Either implement the required logic or convert it to an issue tracker ticket.
...
## File: `lib/screens/parent_module/parent_login/parent_login_screen.dart` (1 issue)

### 88. [CODE_QUALITY - MEDIUM] Unused Widgets/Classes/Functions (Line 327)
- **Evidence:** `// TODO: Navigate to Terms & Conditions`
- **Fix Instruction:**
...
// In lib/screens/parent_module/parent_login/parent_login_screen.dart (Line 327):
// Complete or track pending note: "// TODO: Navigate to Terms & Conditions"
// Either implement the required logic or convert it to an issue tracker ticket.
...

### Fix Rules & Constraints:
1. DO NOT break existing functionality, imports, styles, or public APIs.
2. If packages are required (e.g. `flutter_secure_storage` or `flutter_dotenv`), add them to `pubspec.yaml` / `package.json` appropriately.
3. Ensure no regressions or compilation errors are introduced.
```

8. Analysis Criteria Observed

The project is audited against the following rules:

Code Quality

- **Proper use of const and final:** If widget not updated once declare and variable value will never change once declare then use const and final with declaration (Threshold: `const_possible == true` `assignment_count == 1`)
- **Large File LOC:** Trigger when a file has more than 500 non-empty, non-comment code lines (Threshold: `file_code_loc > 500`)
- **Large Class LOC:** Trigger when a class exceeds configured LOC threshold (Threshold: `class_code_loc > 300`)
- **Large Function LOC:** Trigger when a function/method exceeds configured LOC threshold (Threshold: `function_code_loc > 80`)
- **Unused Widgets/Classes/Functions:** Declared widget/class/function has no references (Threshold: `reference_count == 0`)
- **Unused Imports:** Imported symbol is never referenced (Threshold: `import_reference_count == 0`)
- **Unreachable Branch:** Branch can never execute based on static control-flow analysis (Threshold: `unreachable == true`)
- **High Cyclomatic Complexity:** Function exceeds complexity threshold (Threshold: `complexity > 10`)
- **Duplicate Code:** Similar code block exceeds configured similarity threshold (Threshold: `similarity >= 0.85`)
- **Large Build Method:** Build method exceeds configured size (Threshold: `build_loc > 40`)
- **Deep Widget Nesting:** Widget tree nesting exceeds threshold (Threshold: `depth > 8`)
- **Broad State Rebuild:** State update potentially rebuilds unnecessarily large widget subtree (Threshold: `Configurable`)
- **Missing Const Opportunity:** Constructor/widget can safely be const but is not (Threshold: `const_possible == true`)
- **Improper final Usage:** Local variable never reassigned and can be final (Threshold: `assignment_count == 1`)
- **Unsafe Late Usage:** late variable may be accessed before initialization or is unnecessary (Threshold: `Configurable`)
- **Naming Convention Violation:** Classes, variables, methods/files violate Dart naming conventions (Threshold: `Dart convention`)
- **Empty Catch Block:** Catch block has no meaningful handling/logging/rethrow (Threshold: `catch_body_effective_statements == 0`)

Security

- **Hardcoded API Key:** API key pattern detected in source/config (Threshold: `Pattern + entropy score`)
- **Payment Gateway Credential:** Payment provider private/secret credential detected (Threshold: `Provider pattern`)
- **Firebase Service Account:** Firebase service-account JSON/private key detected (Threshold: `Credential pattern`)
- **Private Key/Certificate:** PEM/private key/signing material detected (Threshold: `private_key_detected == true`)
- **Insecure Token Storage:** Access/refresh tokens stored in SharedPreferences/plain files (Threshold: `Sensitive token + insecure storage`)
- **HTTP/Cleartext Traffic:** Non-TLS HTTP endpoint or cleartext platform configuration detected (Threshold: `http:// / cleartext enabled`)
- **Certificate Validation Bypass:** Client bypasses certificate validation (Threshold: `Trust-all callback/API`)
- **Unsafe WebView Configuration:** JavaScript/file access/untrusted navigation enabled without controls (Threshold: `Risk pattern`)
- **Unsafe Dynamic Data Handling:** Unsafe eval-like/dynamic deserialization or execution pattern detected (Threshold: `Risk API pattern`)
- **Sensitive Data in Logs:** Token/password/payment/PII passed to logging APIs (Threshold: `Sensitive variable/pattern`)
- **Android Insecure Configuration:** Debuggable release config, exported components or permissive network settings (Threshold: `Risk configuration`)
- **iOS Insecure Configuration:** ATS exceptions, insecure entitlements or overly permissive settings (Threshold: `Risk configuration`)
- **Overly Permissive Firebase Rules:** Firestore/Realtime Database allows broad unauthenticated read/write (Threshold: `Broad read/write access`)

Memory Leak

- **Unclosed StreamSubscription:** StreamSubscription created without matching cancellation/disposal (Threshold: `cancel_missing == true`)
- **Unclosed Timer:** Timer created without cancellation (Threshold: `cancel_missing == true`)
- **AnimationController Not Disposed:** AnimationController created in State without dispose (Threshold: `dispose_missing == true`)
- **TextEditingController Not Disposed:** TextEditingController retained without disposal (Threshold: `dispose_missing == true`)
- **Scroll/Page Controller Not Disposed:** ScrollController/PageController created without disposal (Threshold: `dispose_missing == true`)
- **Listener Not Removed:** addListener/addObserver without matching remove (Threshold: `remove_missing == true`)
- **Unnecessary StatefulWidget:** StatefulWidget has no mutable state or lifecycle requirement (Threshold: `stateful_need == false`)
- **Missing Const StatelessWidget:** Immutable widget does not use const constructor (Threshold: `const_possible == true`)
- **Long-Lived Closure Captures State/Context:** Long-lived callback captures State/BuildContext (Threshold: `Risk pattern`)
- **Unbounded In-Memory Collection:** List/map/cache grows without size/eviction control (Threshold: `Configurable size/eviction rule`)

Runtime Analysis

- **Heap Growth Over Time:** Heap increases across repeated flows and does not return toward baseline (Threshold: `Configurable growth %`)
- **Widget/Object Retention:** Objects remain retained after route/widget disposal (Threshold: `Retained after disposal`)
- **Frame Rendering/Jank:** Frame build/raster time exceeds threshold or dropped frames detected (Threshold: `frame_time > 16.67ms`)
- **Excessive Widget Rebuilds:** Widget rebuild count significantly exceeds expected baseline (Threshold: `Configurable multiplier`)
- **Slow Startup:** Cold-start time exceeds configured threshold (Threshold: `startup_ms > 3000`)
- **Slow API Request:** API request exceeds latency threshold (Threshold: `latency_ms > 2000`)
- **Excessive Network Traffic:** Unexpected request frequency or payload volume (Threshold: `Configurable request/payload threshold`)
- **Unhandled Exception:** Unhandled Dart/Flutter exception during test flow (Threshold: `unhandled_exception == true`)
- **Resource Retained After Screen Exit:** Subscription/timer remains active after navigation away (Threshold: `Active resource after dispose`)
- **Sensitive Runtime Storage:** Token/payment/PII written to logs or insecure storage (Threshold: `Sensitive data detected`)

Dependencies

- **Vulnerable Package Version:** Direct/transitive dependency has known security advisory (Threshold: `Known CVE/advisory`)
- **Unused Dependency:** Declared package has no project references (Threshold: `reference_count == 0`)
- **Unmaintained Package:** Package has stale releases or weak maintenance signals (Threshold: `Configurable inactivity period`)

Architecture

- **Layer Violation:** UI directly accesses infrastructure/data layer against selected architecture (Threshold: `Forbidden dependency`)
- **Mixed State Management:** Conflicting state-management patterns used across modules (Threshold: `Configured allowed patterns`)
- **Business Logic Inside Widget:** Networking/business rules embedded in build/UI classes (Threshold: `Configurable complexity/API count`)

Reporting

- **File/Line/Code Evidence:** Every finding must include exact source location and evidence snippet (Threshold: `Required fields`)
- **Severity/Confidence Scoring:** Every finding receives severity and confidence (Threshold: `Required fields`)
- **Suppression/Ignore Support:** Approved findings can be suppressed with a reason (Threshold: `Suppression requires reason`)